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Risk Factors of Ischemic and Hemorrhagic Stroke During Venovenous Extracorporeal Membrane Oxygenation: Analysis of Data from the Extracorporeal Life Support Organization Registry

Sung-Min Cho, DO, MHS¹, Joe Canner, MHS², Giorgio Caturegli, MD¹, Chun Woo Choi, MD², Eric Etchill, MD², Katherine Giuliano, MD², Giovanni Chiarini, MD^{3,4}, Kate Calligy, RN², Peter Rycus, MPH⁵, Roberto Lorusso, MD, PhD³, Bo Soo Kim, MD², Marc Sussman, MD², Jose I. Suarez, MD¹, Romergryko Geocadin, MD¹, Errol L. Bush, MD⁶, Wendy Ziai, MD^{1,*}, Glenn Whitman, MD^{2,*}

¹Division of Neuroscience Critical Care, Departments of Neurology and Anesthesiology and Critical Care Medicine, Johns Hopkins University School of Medicine, Baltimore, Maryland.

²Division of Cardiac Surgery, Cardiovascular Surgical Intensive Care, Department of Surgery, Heart and Vascular Institute, Johns Hopkins University School of Medicine, Baltimore, Maryland.

³Cardio-Thoracic Surgery Department, Maastricht University Medical Centre (MUMC), Cardiovascular Research Institute Maastricht (CARIM), Maastricht, Netherlands.

⁴Division of Anesthesiology, Intensive Care and Emergency Medicine, Spedali Civili University, Affiliated Hospital of Brescia, Brescia, Italy

⁵Extracorporeal Life Support Organization (ELSO), Ann Arbor, Michigan.

⁶Division of Thoracic Surgery, Department of Surgery, Heart and Vascular Institute, Johns Hopkins University School of Medicine, Baltimore, Maryland.

Abstract

Objective: Stroke is commonly reported in patients receiving venovenous extracorporeal membrane oxygenation (V-V ECMO), but risk factors are not well described. We sought to determine pre-ECMO and on-ECMO risk factors for both ischemic and hemorrhagic stroke in patients with V-V ECMO support.

Design: Retrospective analysis.

Setting: Data reported to the Extracorporeal Life Support Organization by 366 ECMO centers from 2013 to 2019.

Patients: Patients older than 18 years old supported with a single run of V-V ECMO.

Address for correspondence and reprints: Sung-Min Cho, DO, MHS, Neuroscience Critical Care Division, Departments of Anesthesiology and Critical Care Medicine, Neurology, and Neurosurgery, The Johns Hopkins School of Medicine, 600 N. Wolfe Street, Phipps 455, Baltimore, MD 21287, Tel: (410)-955-2109, Fax: (410) 614-7903, csungmi1@jhmi.edu.

*W.Z. and G.J.W. contributed equally as senior authors.

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Measurements and Main Results: Of 15,872 V-V ECMO patients, 812 (5.1%) had at least one type of acute brain injury (ABI), defined as ischemic stroke, hemorrhagic stroke, or brain death. Overall, 215 (1.4%) experienced ischemic stroke and 484 (3.1%) experienced hemorrhagic stroke. Overall in-hospital mortality was 36%, but rates were higher in those with ischemic or hemorrhagic stroke (68% and 73%, respectively). In multivariable analysis, pre-ECMO pH (adjusted OR [aOR]=0.10; 95% CI, 0.03–0.35; $p<0.001$), hemolysis (aOR=2.27; 95% CI=1.22–4.24; $p=0.010$), gastrointestinal hemorrhage (aOR=2.01; 95% CI=1.12–3.59; $p=0.019$), and disseminated intravascular coagulation (aOR=3.61; 95% CI=1.51–8.66; $p=0.004$) were independently associated with ischemic stroke. Pre-ECMO pH (aOR=0.28; 95% CI, 0.12–0.65; $p=0.003$), pre-ECMO PO₂ (aOR=0.96; 95% CI=0.93–0.99; $p=0.021$), gastrointestinal hemorrhage (aOR=1.70; 95% CI=1.15–2.51; $p=0.008$), and renal replacement therapy (aOR=1.57; 95% CI=1.22–2.02; $p<0.001$) were independently associated with hemorrhagic stroke.

Conclusions: Among V-V ECMO patients in the ELSO registry, approximately 5% had ABI. Mortality rates increased 2-fold when ischemic or hemorrhagic strokes occurred. Risk factors such as lower pH and hypoxemia during the peri-cannulation period and markers of coagulation disturbances were associated with ABI. Further research on understanding pre-ECMO and on-ECMO risk factors and the timing of ABI is necessary to develop appropriate prevention and management strategies.

Keywords

Extracorporeal membrane oxygenation; venovenous ECMO; ischemic stroke; hemorrhagic stroke; brain injury; brain death

INTRODUCTION

Veno-venous extracorporeal membrane oxygenation (V-V ECMO) is a life-saving intervention that provides pulmonary support for patients with severe acute respiratory failure refractory to mechanical ventilation. The use of V-V ECMO has increased in recent years¹ and has reduced 60-day mortality in patients with acute respiratory distress syndrome (ARDS) when compared to that in patients who receive conventional mechanical ventilation². Prior analysis of Extracorporeal Life Support Organization (ELSO) registry data showed that the presence of neurologic complications or acute brain injury (ABI, defined as ischemic stroke, intracranial hemorrhage, and seizure) increases the risk of poor functional outcome and mortality from ECMO³. In that study, 7% of adults with V-V ECMO experienced neurologic complications, with an associated mortality of 80% for intracranial hemorrhage and 68% for ischemic stroke³.

Stroke is often perceived to be less common in patients who receive V-V ECMO than in those who receive veno-arterial (V-A) ECMO, as the V-V ECMO circuit does not involve the arterial circulation and V-V ECMO is often run with less or even without anticoagulation, compared to V-A ECMO. However, in a recent publication, ABI appears to occur with equal frequency regardless of ECMO type⁴. Furthermore, when V-V ECMO patients underwent routine computed tomography (CT) scans within 24 hours of cannulation, 16%

had hemorrhagic strokes⁵. In the population who underwent postmortem neuropathologic evaluation, 68% of ECMO non-survivors exhibited ABI, suggesting that the true prevalence of ABI is likely higher than that previously reported⁶. Currently, there is a large gap in knowledge regarding the prevalence, risk factors, timing, and management of ECMO-related ABI^{4,7}. Understanding mechanisms and risk factors of ischemic and hemorrhagic stroke in patients receiving V-V ECMO will provide crucial information to guide prevention and inform appropriate management strategies.

The aim of this study was to identify risk factors for ischemic and hemorrhagic stroke in patients receiving V-V ECMO support in an effort to provide an understanding of potential mechanisms for each stroke subtype. We hypothesized that ischemic stroke and hemorrhagic stroke have different pre-ECMO and on-ECMO risk factors.

Materials and Methods

Study Design and Population

The ELSO registry is a voluntary database that collects information on use, indications, complications, and outcomes of ECMO support in adults and children from 366 member centers worldwide⁸. Records stored in the registry include patient demographics; clinical characteristics; pre-ECMO conditions, including pulmonary diagnoses; hemodynamic and laboratory values collected before and during ECMO; complications that occur during ECMO support, including neurologic complications; and outcome data such as survival to hospital discharge. Diagnosis and medical history are reported according to the International Classification of Diseases (ICD) 9th edition (ICD-9) and 10th (ICD-10) edition codes.

We carried out a retrospective analysis of the ELSO registry database from 2013 through 2019. We included only patients 18 years and older who received one run of V-V ECMO support for respiratory failure. We excluded patients who underwent multiple runs to avoid complexity and bias in the analysis resulting from severe illness, as we considered that the morbidity during the inter-ECMO periods could introduce significant confounders. Patients treated with V-A ECMO were excluded.

Data Collection and Definitions

For all patients in the study, we extracted the following information from the ELSO registry database: pre-ECMO demographic information; pre-ECMO clinical variables; cardiac diagnoses; pulmonary diagnoses; laboratory values; on-ECMO clinical variables, including hemodynamic data and the use of other supportive cardiac devices; and ECMO-associated morbidity and mortality, including neurologic complications. The pre-extracorporeal life support (ECLS) hemodynamics and arterial blood gas (ABG) values were measured no more than 6 hours before initiation of ECLS; if the database contained multiple values, we chose those closest to the time of cannulation. Twenty-four-hour ABG values were drawn between 18 and 30 hours after ECLS start time. The “ECMO circuit mechanical failure” variable included oxygenator failure, cracks in pigtail connectors, clots in the hemofilter, clots in the circuit component, circuit change, cannula problems, air in the circuit, pump failure, and tubing rupture.

Neurologic complications included ischemic stroke, hemorrhagic stroke, brain death, and seizure. In the ELSO database, ischemic stroke is defined as central nervous system (CNS) infarction determined by ultrasound, CT, or magnetic resonance imaging (MRI). Hemorrhagic stroke is defined as intra- or extra-parenchymal CNS hemorrhage determined by ultrasound, CT, or MRI. The ELSO database has two seizure variables: those determined by clinical assessment and those confirmed by electroencephalogram. We combined these two variables as “seizure” for the analysis. Although new neurologic variables, including “CNS diffuse ischemia,” “intraventricular hemorrhage,” and “neurosurgical intervention performed,” recently have been added to the ELSO database addendum, these were not available in the current analysis. Additionally, subtypes of hemorrhagic stroke such as intracerebral hemorrhage, subarachnoid hemorrhage, and subdural hematoma were not available in the registry. This study was reviewed by the Johns Hopkins Institutional Review Board (IRB) and waived the need for approval.

Statistical Analysis

Demographic and clinical variables in subjects with and without each type of neurologic injury (ischemic stroke, hemorrhagic stroke, and brain death) were compared by unpaired t-test, chi-squared test, or Mann-Whitney U test as appropriate. Results are expressed as the mean with standard deviation if normally distributed or as the median with quartiles if non-normally distributed for quantitative variables and as proportions for categorical findings. Ordinal or categorical data are reported as numbers and percentages. A p value <0.05 was considered statistically significant. Odds ratios (ORs) with 95% confidence intervals (CIs) were calculated by logistic regression analysis. We carefully selected clinically relevant variables that had biologically plausible associations or causality for inclusion in multivariable logistic regression analysis to identify risk factors that were associated with each of ischemic and hemorrhagic stroke compared to patients with no known stroke. Akaike information criterion was calculated for selecting the best model. All analyses were carried out in STATA 15 (College Station, TX, USA).

RESULTS

A total of 15,872 patients (median age 49 years, 61% male) who received V-V ECMO support were included in the study. The use of V-V ECMO increased over time 1,179 in 2013 to 3,011 in 2019, representing a 2.5-fold increase in 7 years (Table 1). Of the 15,872 patients, 812 (5.1%) had at least one type of ABI, including ischemic stroke, hemorrhagic stroke, or brain death (Fig. 1). Overall, 215 (1.4%) patients had ischemic stroke and 484 (3.1%) had hemorrhagic stroke. Brain death was reported in 204 (1.3%) patients. Although the rates of V-V ECMO-associated ischemic stroke remained stable over 7 years ($p=0.18$), the rates of hemorrhagic stroke and brain death decreased over time ($p<0.001$ and $p=0.002$, respectively; Table 1 and Supplemental Table 1). Overall, ABI during V-V ECMO decreased from 6.2% in 2013 to 5.1% in 2019 (Fig. 1). The mortality for all ECMO patients was 36% over 7 years, a rate that increased to 68% in the presence of ischemic stroke and 73% in the presence of hemorrhagic stroke. Overall mortality remained stable over time from 38% to 36% as did mortality due to ABI (77% to 74%) during the study period.

Associations with ABI: Univariate analysis

Among the 215 patients who had ischemic strokes, the median age was 50 and 59% were male. Compared to patients with no stroke, patients who exhibited a higher pre-ECMO PCO₂, lower pre-ECMO pH, PO₂, and mean arterial pressure (MAP) were more likely to have ischemic stroke. During ECMO support, higher rates of the following ECMO complications were associated with ischemic stroke: renal replacement therapy (RRT), hypoglycemia, hemolysis, cardiac arrhythmia, gastrointestinal hemorrhage, disseminated intravascular coagulation (DIC), and seizure (Table 1).

Compared to patients without stroke, the 484 patients with hemorrhagic stroke (median age 49 years, 56% male) had lower weight, lower pre-ECMO PO₂, higher pre-ECMO PCO₂, lower pH and PO₂ on ECMO, longer duration of ECMO support, and higher rates of ECMO complications including RRT, hypoglycemia, hemolysis, cardiac arrhythmia, gastrointestinal hemorrhage, DIC, and seizure (Table 1).

Compared to patients without brain death, patients with brain death were younger, had lower pre-ECMO pH and HCO₃, higher pre-ECMO PCO₂, and lower pre-ECMO blood pressure values. They also had a lower pH and HCO₃ on ECMO, higher pCO₂ on ECMO, shorter duration of ECMO support, and a higher incidence of ECMO complications including RRT, DIC and seizure, than did patients without brain death (Supplemental Table 1).

Pre-ECMO PO₂ values were lower among those who had ischemic stroke than among those without stroke (57 vs. 63 mmHg, $p=0.002$). Similarly, pre-ECMO PO₂ values were lower in patients with hemorrhagic stroke than in those without (60 vs. 63 mmHg, $p<0.001$). Supplemental Figure 1 demonstrates that there are more patients with hypoxemia in the hemorrhagic stroke group vs. without (B). Even at 24 hours of ECMO support, lower PO₂ values were observed in patients with hemorrhagic stroke (74 vs. 80 mmHg, $p<0.001$).

Associations with ABI: Multivariable analysis

In multivariable analysis, pre-ECMO pH (adjusted OR [aOR]=0.10; 95% CI, 0.03–0.35; $p<0.001$), hemolysis (aOR=2.27; 95% CI=1.22–4.24; $p=0.010$), gastrointestinal hemorrhage (aOR=2.01; 95% CI=1.12–3.59; $p=0.019$), and disseminated intravascular coagulation (aOR=3.61; 95% CI=1.51–8.66; $p=0.004$) were independently associated with ischemic stroke (Table 2). Pre-ECMO pH (aOR=0.28; 95% CI, 0.12–0.65; $p=0.003$), pre-ECMO PO₂ (aOR=0.96; 95% CI=0.93–0.99; $p=0.021$), gastrointestinal hemorrhage (aOR=1.70; 95% CI=1.15–2.51; $p=0.008$), and renal replacement therapy (aOR=1.57; 95% CI=1.22–2.02; $p<0.001$) were independently associated with in hemorrhagic stroke (Table 3).

DISCUSSION

Incidence and Mortality

Our analysis of 15,872 V-V ECMO patients in the ELSO registry showed that approximately 5% had at least one type of ABI, defined as ischemic stroke, hemorrhagic stroke, or brain death. During the study period, reported ABI rates during V-V ECMO remained stable at 5–6%. Although mortality was 36% overall in V-V ECMO patients, not surprisingly, the

mortality rate was doubled when ischemic stroke or hemorrhagic stroke was present, as reported previously^{3,4,7}. Mortality associated with ischemic or hemorrhagic stroke remained stable over 7 years. Notably, the observed rates of ischemic stroke and hemorrhagic stroke in the ELSO database were lower (at 1.4% and 3.1%, respectively) than those reported in prior literature (4–14% for ischemic stroke; 4–19% for hemorrhagic stroke)^{4,5,9–13}. However, this perceived low rate of stroke determined by our study is likely an underestimation because ELSO data is reported voluntarily without central or local adjudication of neurological diagnoses, and CT scans of the brain were not mandated to make a stroke diagnosis in the registry. Furthermore, CT scans are insensitive for detecting early cerebral ischemic changes and posterior fossa ischemia. More dramatic was the finding in two recent brain autopsy studies that 69–90% of ECMO non-survivors had ABI on postmortem pathological evaluation^{6,14}, suggesting the true prevalence of ABI is likely much higher than that reported in clinical observational studies. This underscores the importance of protocolized neuromonitoring and systematic neurological care in these patients if we are to better understand the neurologic sequelae of ECMO, and improve timely diagnoses so as to provide appropriate therapeutic interventions for these patients^{7,15,16}.

Ischemic Stroke

Risk factors and mechanisms of ischemic stroke for patients on V-V ECMO are ill-defined^{3,9,10}. Critical illness is strongly associated with cardiac arrhythmias such as atrial fibrillation for example, a known cause for embolic strokes, as well as hypercoagulability and the difficulty in managing anticoagulation therapy^{4,17,18}. Our results showed that cardiac arrhythmias were not an independent risk factor for ischemic stroke. Other biologically plausible causes of V-V ECMO-associated ischemic stroke include emboli from the circuit through a patent foramen ovale as well as cerebral venous sinus thrombosis (CVST)⁴. The use of the internal jugular vein for cannulation is a known risk factor for CVST if thrombosis extends intracranially, a risk that may be increased with the use of dual-lumen right internal jugular veno-venous catheters, currently in common use, but with significantly larger internal diameter than the single lumen cannulae used routinely in the past⁴. Interestingly, when CT scans were performed within 4 days after V-V ECMO femoro-jugular decannulation, 53% (42 of 79) of patients were found to have cannula-associated jugular deep vein thrombosis¹⁹. The ELSO database does not collect information on CVST, and thus, we could not evaluate its association with ABI.

We identified four independent variables that were associated with ischemic stroke, namely, lower pre-ECMO pH and three on-ECMO complications, hemolysis, gastrointestinal hemorrhage, and DIC. These findings may provide insight into the timing and type of ischemic stroke in patients receiving V-V ECMO. Lower pre-ECMO pH in the ischemic stroke group suggests that severe refractory respiratory failure along with hemodynamic instability before cannulation may be a cause of ischemic stroke. The ELSO registry did not differentiate stroke subtypes, limiting our ability to distinguish global brain ischemia, leading to hypoxic-induced brain injury (HIBI) or watershed infarcts from other stroke subtypes as defined by TOAST criteria (Trial of Org 10172 in Acute Stroke Treatment) including large-artery atherosclerosis, small-vessel occlusion, and cardioembolism²⁰. While pre-ECMO hypoxic respiratory failure may predispose to global brain ischemia, hemolysis

and DIC, independent “on-ECMO” risk factors, may lead directly to thromboembolic stroke given its associated microangiopathy²¹. Notably, hemolysis and subclinical pump thrombosis, (which could cause elevations in plasma free hemoglobin that we observed) are known risk factors for ischemic infarct in the LVAD population²². In contrast, gastrointestinal hemorrhage is more likely a surrogate for anticoagulation disruption or may represent severity of critical illness with ischemic stroke causing stress ulceration.

Hemorrhagic Stroke

We found four independent variables that were associated with hemorrhagic stroke including lower pre-ECMO pH and PO₂ and two on-ECMO complications, RRT and gastrointestinal hemorrhage (Table 3). Mechanical circulatory support devices such as LVAD and ECMO cause platelet dysfunction and endothelial cell dysfunction with subsequent acquired von Willebrand syndrome (Factor VIII deficiency), a known risk factor for bleeding^{24–26}. Our finding of RRT increasing hemorrhage risk is consistent with prior studies reporting thrombocytopenia^{27,28} and renal failure^{4,9,12} as risk factors for hemorrhagic stroke. We also observed that both pre- ECMO pH and PO₂ values were lower in the hemorrhagic stroke group (Table 1, Supplemental Figure 1). The absolute difference in median values of pH and PO₂ between the groups (0.02 in pH and 3 mmHg in PO₂, Table 1) is unlikely to be clinically meaningful. However, one might hypothesize that hemorrhagic stroke occurs in the peri-ECMO cannulation period with refractory hypoxia and hemodynamic instability. A higher number of RRT also supports worse hemodynamic instability in the hemorrhagic stroke group. This conjecture is supported by recent observational studies in which consecutive V-V ECMO patients received early routine CT scans, which reported a 16% hemorrhagic stroke rate⁵ and 10% rate in those receiving extracorporeal cardiopulmonary resuscitation (eCPR)²⁹. Given these recent data on “early” hemorrhagic stroke in ECMO, we suggest two hypotheses on potential mechanisms of hemorrhagic stroke in V-V ECMO. First, early hemorrhagic stroke may occur secondary to hemorrhagic conversion of “fresh” hypoxic-ischemic brain tissue³⁰ from refractory respiratory failure with the use of bolus dose heparin during ECMO cannulation and then maintenance anticoagulation during the ECMO run. Second, hemorrhagic stroke may occur in patients with severe ARDS independent of ECMO initiation. Lockie et al.⁵ reported that ~8% of ARDS patients requiring mechanical ventilation had hemorrhagic stroke without ever being exposed to heparin or the ECMO circuit. However, the incidence of hemorrhagic stroke was twice that (16%) among those treated with V-V ECMO⁵. Similarly, patients with critical illness and severe respiratory failure may suffer from cerebral microhemorrhages^{31,32}; thus, many hemorrhagic strokes may arguably precede the ECMO run. Additional research is needed to understand the association between severe hypoxic respiratory failure and critical illness leading to cerebral micro- and macro-hemorrhage and its association with anticoagulation intensity.

Limitations

The main limitations of our study were that it was a retrospective analysis and the ELSO registry lacks several important explanatory variables for the diagnoses of ischemic and hemorrhagic stroke including anticoagulation data, occurrence of atrial fibrillation, patent foramen ovale, platelet counts, and other stroke risk factors. We were also unable to measure

exposure duration to hypoxemia/acidosis as we had only one ABG data point for each pre- and on-ECMO support. The incidence of ABI in the ELSO registry likely is an underestimation as entries are reported voluntarily without central or local adjudication of neurological diagnosis. Also, it is unclear when and for which indications CT brain studies were performed and the criteria followed for brain death. The findings of our multivariable logistic regression models for ABI can only describe associations and do not imply these factors are predictive, owing to the retrospective design and uncertainty regarding the timing of strokes and covariates. The clinical relevance of some small differences in physiologic variables such as PO₂ and pH requires further evaluation and may have occurred by chance due to the high number of values analyzed. Strength of our study was its large sample size and high statistical power. The ELSO registry is the largest international repository of ECMO data, offering a distinctive opportunity to evaluate the incidence, mortality, and risk factors of each type of ABI in a large number of patients.

Conclusions

Among V-V ECMO patients in the ELSO registry, approximately 5% experienced ABI. Mortality rates were high (~70%) when ischemic and hemorrhagic strokes were present compared to those with no strokes. As the use of ECMO is increasing exponentially, future research should be directed toward use of neurologic monitoring for early recognition and development of targeted management strategies especially with regard to timing and intensity of anticoagulation.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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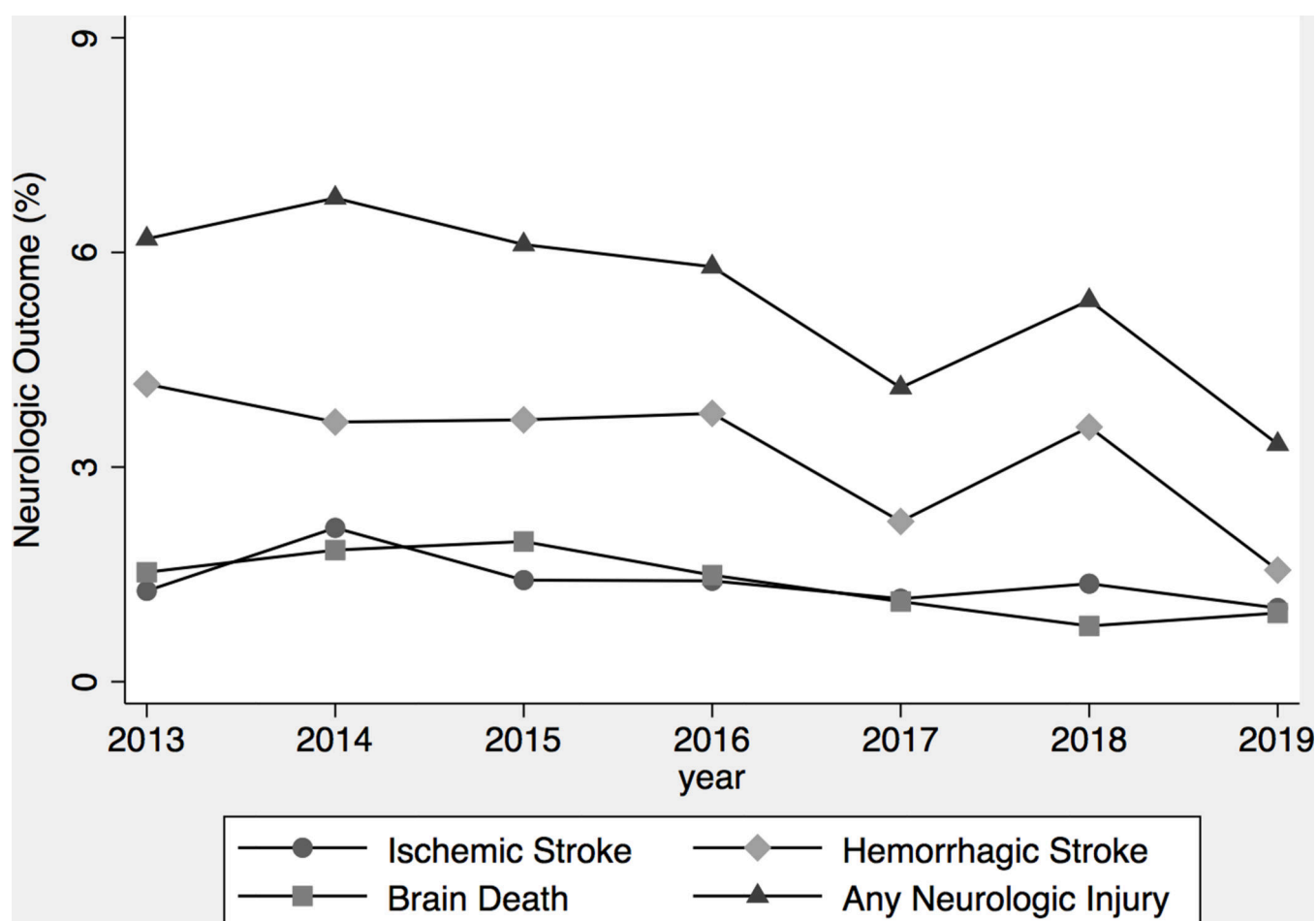


Figure 1. The incidence of acute brain injury in patients treated with venovenous extracorporeal membrane oxygenation, 2013–2019. Data were collected from the Extracorporeal Life Support Organization registry.

Table 1.

Baseline demographics and ECMO variables in V-V ECMO patients.

Variables	Without any stroke (n=15,206)	Ischemic stroke (n=215)	P-value	Hemorrhagic stroke (n=484)	P-value
Age (years), median (IQR) *	48 (35–60)	50 (36–61)	0.19	49 (37–59)	0.40
Weight (kg), median (IQR) *	85 (70–103)	85 (72–100)	0.75	80 (66–100)	0.002
Male *	9,108 (61)	125 (59)	0.46	271 (56)	0.03
Race *					
White	8,633 (58)	132 (63)		277 (58)	
Asian	1,841 (12)	26 (12)		90 (19)	
Black	1,710 (11)	17 (8)	0.009	29 (6)	<0.001
Hispanic	1,027 (7)	20 (10)		41 (9)	
Others	1,667 (11)	15 (7)		38 (14)	
Year ECLS					
2013	1,117 (7)	15 (7)		49 (10)	
2014	1,537 (10)	35 (16)		59 (12)	
2015	1,746 (11)	26 (12)		67 (14)	
2016	2,361 (16)	35 (16)	0.07	93 (19)	<0.001
2017	2,594 (17)	31 (14)		60 (12)	
2018	2,915 (19)	42 (20)		109 (23)	
2019	2,936 (19)	31 (14)		47 (10)	
ECMO indication					
Acute respiratory distress syndrome	2,534 (17)	41 (19)	0.35	94 (19)	0.11
Influenza	931 (6)	12 (6)	0.74	60 (12)	<0.001
Viral Pneumonia	465 (3)	7 (3)	0.87	25 (5)	0.009
Bacterial Pneumonia	678 (4)	9 (4)	0.85	30 (6)	0.07
Chronic obstructive pulmonary disease	479 (3)	5 (2)	0.49	4 (1)	0.004
Asthma	387 (3)	4 (2)	0.53	3 (1)	0.007
Pulmonary edema	20 (0.1)	0 (0)	0.60	0	0.43
Pulmonary embolism	93 (1)	5 (2)	0.002	2 (0.4)	0.58
Acute respiratory failure	3,256 (21)	49 (23)	0.63	79 (16)	0.007

Variables	Without any stroke (n=15,206)	Ischemic stroke (n=215)	P-value	Hemorrhagic stroke (n=484)	P-value
Pneumonitis	175 (1)	1 (0.5)	0.35	3 (1)	0.28
Pre-ECMO ABG					
pH ^{*o}	7.24 (7.14–7.33)	7.20 (7.09–7.28)	<0.001	7.22 (7.14–7.30)	0.002
pCO ₂ , mmHg ^{*o}	58 (47–74)	64 (49–79)	0.01	62 (51–80)	<0.001
pO ₂ , mmHg ^{*o}	63 (50–80)	57 (44–79)	0.002	60 (48–75)	<0.001
HCO ₃ , mmol/L [*]	24.0 (20.0–29.0)	23.7 (20.0–28.9)	0.37	24.7 (21.0–29.5)	0.08
Mixed venous oxygen saturation [*]	66 (55–77)	60 (51–76)	0.46	67 (58–78)	0.90
Cardiac index, (L/min/m ²) [*]	3.0 (2.3–3.9)	3.2 (2.8–4.3)	0.44	2.8 (2.2–4.5)	0.87
Pre-ECMO hemodynamics					
Systolic blood pressure, mmHg [*]	106 (90–125)	103 (88–125)	0.43	105 (90–122)	0.40
Diastolic blood pressure, mmHg [*]	58 (50–68)	57 (48–67)	0.23	57 (48–66)	0.07
Mean blood pressure, mmHg [*]	72 (62–84)	68 (59–82)	0.03	71 (62–81)	0.37
ABG at 24 hours					
pH ^{*o}	7.40 (7.35–7.44)	7.40 (7.35–7.44)	0.75	7.40 (7.35–7.44)	0.91
pCO ₂ , mmHg ^{*o}	41 (37–46)	41 (36–46)	0.41	42 (37–46)	0.22
pO ₂ , mmHg ^{*o}	80 (65–108)	81 (61–107)	0.58	74 (60–96)	<0.001
HCO ₃ , mmol/L [*]	25.0 (22.0–28.2)	24.2 (22.0–28.0)	0.14	24.9 (22.0–28.9)	0.72
Days on ECMO support [*]	7.8 (4.0–14.1)	7.6 (3.3–15.7)	0.66	8.6 (4.0–15.8)	0.037
ECMO complications					
ECMO circuit mechanical failure	3,398 (22)	55 (26)	0.26	107 (22)	0.90
Renal replacement therapy	4,171 (27)	86 (40)	<0.001	204 (42)	<0.001
Glucose < 40 mg/L	142 (1)	5 (2)	0.037	11 (2)	0.003
Hemolysis (plasma hemoglobin >50 mg/dL)	573 (4)	19 (9)	<0.001	31 (6)	0.003
Cardiac arrhythmia	1,068 (8)	25 (12)	0.009	62 (13)	<0.001
Gastrointestinal hemorrhage	721 (5)	23 (11)	<0.001	46 (10)	<0.001
Disseminated intravascular coagulation	205 (1)	10 (5)	<0.001	12 (2)	0.036
Seizure	146 (1)	15 (7)	<0.001	17 (4)	<0.001

All data are presented as n (%) for categorical variables and median (interquartile range) for continuous variables.

ECMO = extracorporeal membrane oxygenation; IQR = interquartile range; V-V = veno-venous; ABG = arterial blood gas.

Bold text indicates statistical significance.

* Missing values for some variables resulted in different denominators for the following variables: age (n=15,814), weight (n=15,092), sex (n=15,367), race (n=15,530), pH (n=13,284), pCO₂ (n=12,294), pO₂ (n=13,017), HCO₃ (n=12,444), mixed venous oxygen saturation (n=1,227), cardiac index (n=680), systolic blood pressure (n=11,719), diastolic blood pressure (n=11,661), mean blood pressure (n=10,627), 24 hour pH (n=14,022), 24 hour pCO₂ (13,017), 24 hour pO₂ (n=13,695), 24 hour HCO₃ (n=13,762) days on ECMO support (n=15,742).

° The following values were excluded due to biological implausibility: pH>8, pH<6.5, pO₂>760 mmHg, pCO₂>240 mmHg, pCO₂<20 mmHg.

Table 2.

Multivariable analysis of ischemic stroke in patients with V-V ECMO

Ischemic stroke	Odds Ratio	95% C.I.	P-value
<i>Pre-ECMO</i>			
pH	0.10	0.03–0.35	<0.001
pO ₂ (per 10 mmHg)	1.01	0.99–1.04	0.311
Mean blood pressure, mmHg	1.00	0.99–1.01	0.878
<i>ECMO course</i>			
pCO ₂ at 24 hours (per 10 mmHg)	1.03	0.90–1.17	0.664
pO ₂ at 24 hours (per 10 mmHg)	1.02	0.99–1.04	0.214
HCO ₃ at 24 hours (per 10 mmHg)	1.00	0.96–1.04	0.937
Renal replacement therapy	1.26	0.83–1.84	0.296
Glucose < 40 mg/L	0.73	0.10–5.53	0.764
Hemolysis (plasma hemoglobin >50 mg/dL)	2.27	1.22–4.24	0.010
Cardiac arrhythmia	0.68	0.33–1.43	0.312
Gastrointestinal hemorrhage	2.01	1.12–3.59	0.019
Disseminated intravascular coagulation	3.61	1.51–8.66	0.004

VV = veno-venous; ECMO = extracorporeal membrane oxygenation; C.I. = Confidence Interval.

Bold text indicates statistical significance.

Table 3.

Multivariable analysis of hemorrhagic stroke in patients with V-V ECMO

Hemorrhagic stroke	Odds Ratio	95% C.I.	P-value
<i>Pre-ECMO</i>			
pH	0.28	0.12–0.65	0.003
pO ₂ (per 10 mmHg)	0.96	0.93–0.99	0.021
Mean blood pressure, mmHg	1.00	0.99–1.01	0.549
<i>ECMO course</i>			
pCO ₂ at 24 hours (per 10 mmHg)	1.06	0.92–1.22	0.438
pO ₂ at 24 hours (per 10 mmHg)	0.99	0.96–1.01	0.280
HCO ₃ at 24 hours (per 10 mmHg)	1.02	0.99–1.04	0.204
Renal replacement therapy	1.57	1.22–2.02	<0.001
Glucose < 40 mg/L	2.12	0.83–5.41	0.116
Hemolysis (plasma hemoglobin >50 mg/dL)	1.08	0.64–1.82	0.776
Cardiac arrhythmia	1.11	0.74–1.65	0.612
Gastrointestinal hemorrhage	1.70	1.15–2.51	0.008

V-V = veno-venous; ECMO = extracorporeal membrane oxygenation; C.I. = Confidence Interval.

Bold text indicates statistical significance.